Sakib Raihan

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Summary

AI and ML enthusiast person with a strong foundation in Python. Have hands-on experience in LLMs, Transformers, Vision Models, and RAG systems. Proficient in PyTorch, TensorFlow, and the Python data stack, with full-stack capabilities in FastAPI, Flask, and React. Skilled in fine-tuning, agentic workflows, and deploying models to production. Eager to build impactful AI solutions in a fast-paced environment.

Skills

Programming Languages: Python, SQL, C
Libraries and Tools: PyTorch, Tensorflow, Keras, Sklearn, OpenCV, Pandas, Matplotlib, Numpy, LangChain, LangGraph, Roboflow, Unsloth
AI Expertise: Large Language Models (GPT, LLaMA etc.), Transformers, Vision Models (CNNs, ViT etc.), RAG, Vector Search, Agentic Systems, Prompt Engineering, Fine-tuning & Custom Model Training
Cloud & DevOps: AWS, Docker, Kubernetes, Git, GitHub Actions (CI/CD), System Monitoring
Web Development: Flask, React, Typescript, FastAPI, HTML, CSS
Data Visualization Tools: Power BI
Analytical Abilities: Data Cleaning, Exploratory Data Analysis (EDA), Insight Generation

Experience

AI Engineer Trainee **June 2026 – Present**
Chowa Giken Corporation *Tools Used: CloudCompare, PyTorch etc.*

- Working with **real-world 3D point cloud datasets**, including preprocessing, annotation, feature extraction, and optimization for deep learning and computer vision applications.
- Engaged in industry-oriented on-the-job training in real-world AI development, machine learning workflows, and Japanese engineering standards.

Projects

BRACU BOT | *NLP, RAG, Gradio, FAISS* **May 2025 – June 2025**

- Built an AI chatbot using RAG with SentenceTransformers + FAISS for semantic search and context-aware Q&A on BRAC University content.
- Developed an NLP pipeline for document chunking, vector embeddings, top-k retrieval, and answer generation using Hugging Face Transformers (FLAN-T5).
- Implemented automated web scraping/data ingestion with Playwright + BeautifulSoup to create a structured knowledge base from university webpages.
- Deployed an interactive Gradio interface for real-time responses with a modular architecture optimized for scalability and future feature integration.

Quite Space | *Flask, React, SQLite, NLP (Full-Stack)* **August 2026 – September 2026**

- Implemented intelligent reporting workflows for general and emergency reports with ML-based sentiment analysis, searchable history, and attachment support.
- Developed core group management features including private/public groups, invite codes, member roles, announcements, and real-time chat with file uploads and message history.
- Integrated Flask REST APIs with a structured SQLite data layer to manage users, groups, messages, reports, and notifications across the application.

Multi-Target Domain Adaptation for Semi-Supervised Object Detection | **December 2025 – January 2026**

- Implemented YOLOv11s-based multi-domain object detection framework (normal, foggy, rainy) using PyTorch with adversarial domain classifier and MMD feature alignment, achieving 57% balanced mAP across all domains.
- Built a 3-stage dynamic curriculum learning scheduler that progressively increased domain difficulty, improving rainy-domain detection by 90% (30% → 57% mAP) through adaptive pseudo-labeling.
- Developed semi-supervised pseudo-labeling pipeline with confidence-based filtering (0.6→0.4→0.2) and progressive data inclusion, enabling robust learning from unlabeled target domain data.

HireSkillz | *Flask, React, PostgreSQL, Scikit-learn (Full-Stack)* **February 2026 – March 2026**

- Engineered full-stack career intelligence platform serving 8+ AI-powered modules including NLP-based job matching using TF-IDF vectorization and cosine similarity to connect users with roles, achieving 85% match accuracy across 100+ mock job postings from 20 major tech companies.
- Built ML-driven career recommendation system leveraging scikit-learn feature extraction to match user profiles against 16 career paths with 100+ curated courses, generating personalized recommendations with match percentage scoring for career path pivots.
- Implemented multi-source job aggregation service with real-time scraping, filter pipelines, and normalization across 14 job sources, enabling searchable listings by role, location, work type, and experience level for faster discovery.

Government Service Navigation Agent | *LangGraph, FAISS, LLM* **August 2026 – September 2026**

- Built a bilingual Retrieval Augmented Generation (RAG) AI agent for Bangladeshi government services covering fees, documents and procedures, using a five stage LangGraph pipeline, FAISS vector retrieval, multilingual sentence transformer embeddings, a constrained LLM layer, and automatic language detection.
- Reverse engineered two Bengali PDF text extraction decoders, recovering a corrupted ToUnicode CMap via embedded font glyph names and GSUB closure, and mapping the legacy Bijoy/SutonnyMJ encoding, making government gazettes readable where standard extractors failed.
- Achieved 0% hallucination across a 61 case automated evaluation harness by validating every LLM generated number against source records and rendering all facts deterministically from a validated JSON Schema, with hard grounding and safety gates failing the build on any violation.

Awards & Certifications

- **GCI World Japan Tour (2026):** Participated in a fully funded Japan Study Tour featuring AI research activities at the University of Tokyo and visits to Toyota Tsusho, TOPPAN Holdings, Wanderlust Inc., and the Ministry of Foreign Affairs (MOFA), gaining valuable insights into Japan's technology industry. [View]
- **GCI World Program (2026):** Secured 1st Place in the GCI World Program 2026 Hackathon and was selected as a GCI World 2026 Outstanding Student (Top 17 of 25,000+ participants worldwide), earning an invitation to the Japan Study Tour sponsored by the Matsuo-Iwasawa Laboratory, The University of Tokyo. [View]
- **Nippon AI Dojo Program (2026):** Selected as one of the Top 10 participants from BRAC University for the Nippon AI Dojo 2026 Program, a prestigious AI training initiative organized by CHOWA GIKEN Corporation, Japan, and subsidized by the Ministry of Economy, Trade and Industry (METI), Japan, from a competitive pool of approximately 650 applicants.
- **HerWILL AI for Digital Safety Datathon (2026):** Ranked among the Top 7 Finalists with participants from 45+ countries [View]
- **BAIC Datathon BUBT (2025):** Secured 14th rank among 49 participants from universities across Bangladesh [View]
- **Next Gen Ideathon (2025):** Finalist for developing an AI-driven tourism solution to enhance Bangladesh's travel industry [View]

Education

B.Sc in Computer Science and Engineering **September 2022 - Present**
BRAC University, Dhaka, Bangladesh *CGPA: 3.51/4*

Higher Secondary Certificate **July 2019 - January 2022**
Savar Cantonment Public School and College, Savar, Dhaka, Bangladesh *GPA: 4.58/5*